

Capstone 3: Analysis with Power BI

For this project, you will clean, model, analyze, and visualize business-related data using Power BI.

Requirements

Your capstone project must include **all** of the following to be considered complete:

- Power BI report analyzing the provided sales data, including:
 - A line chart showing sales trends over the full period covered in the data
 - A column or bar chart showing sales performance by category
 - A pie or donut chart showing relative sales for each state in the region
 - A list of top-selling books, including author names, for your region (general audience, excluding textbooks)
- Report must consist of at least two pages, and each page must contain a descriptive header
- All project material published in your Capstone_3 repository by 8:30am Friday, with at least one commit per day to demonstrate your progress throughout the week
- A short README file in your Capstone_3 repository to describe the project
- 10-minute video recording to demonstrate your project and results, with a link to the recording in your Capstone_3 README file
- 5-minute in-class presentation to share your report and key insights
- Final report must be published online to the class workspace in Power BI service

Once again, you will also participate in peer review of your classmates' projects including review of their video recordings and project files.

Note: You will have plenty of opportunities in this project to keep you busy and stretch your skills. Just make sure the core requirements are met first! (See last page for bonus challenges.)

Scenario

You are continuing in your new role as a data analyst for the growing business EmporiUm, which describes itself as a “virtual student bookstore.” EmporiUm got its start in 2022 following the merger of a chain of college bookstore franchises with an online retailer that sold books and tech gear to students. Today, EmporiUm continues to do a blend of in-store and online sales.

Your team has been asked by one of the regional directors for EmporiUm to analyze sales performance over the last four years. You must create a Power BI report that highlights key insights, including trends over time and insights into recent sales highlights, to be shared with the regional sales team (including territory managers and individual store managers). You must also identify top-selling books and their authors, as well as any other insights that could be useful. You will roll out this report at the management team meeting for your region.

Project Setup

1. The capstone project files include a large Excel file containing most of the data, plus a separate text file with book titles and author names for “general” bookstore books (not textbooks). Take a little time to review these files and familiarize yourself with the content and shape of the data, then add these files to your Power BI project.
 - * Make sure to frequently save your project as you work throughout the week!
2. In GitHub, create a new repository named Capstone_3, and clone it to your DATA_Labs local folder.
3. Create a new Power BI report named **YourLastName_capstone3.pbix** and save it to Capstone_3.
4. You will need to start by cleaning and shaping the data into a workable data model.
 - * There are a lot of quirks in this data! Focus your cleaning and transformation on the practical steps to get your data model into shape to support your report visualizations.
 - * You may find it helpful to do some diagramming (or sketching on scrap paper) to work through your ideas on how to reshape the data. What is important? What do you need to support your analysis? This is *not* a required deliverable for your project, but it is recommended!
 - * It is very likely that as you continue working on your report, you will discover additional cleaning steps or other data transformations that you will need to go back into Power Query to work on. This is a normal part of working with data! Do not expect to get everything “perfect” the first time.

Core Project Requirements

- Power BI Report Requirements -

5. Your report must consist of at least two pages, with text headers for each page. You are welcome to customize the theme and formatting of your report however you choose.
6. Your report must include *at least* four visuals:
 - * line chart showing sales trends over the full period covered in the data
 - * column or bar chart showing sales performance by category
 - * pie or donut chart showing relative sales for each state in the region
 - * list of top-selling books, including author names, for your sales region (excluding textbooks)
7. Consider how this information can be used by the sales team to help them make better business decisions. You will need to pitch this to the sales team in your in-class presentation.

8. Remember to save changes as you work throughout the week and commit to your Capstone_3 repo.
9. By Friday morning, use the Publish feature in Power BI to publish the final version of your report to the shared workspace created for your class.

- Capstone Video -

10. Record a 10-minute video to present your capstone project and the process you took. In your video:
 - * Describe what you found in your analysis of the data, going through each of the four required charts one at a time. Include at least one filter interaction to highlight information during your report demonstration.
 - * Include a brief discussion of how you approached cleaning and modeling the data – e.g. what priorities you identified, what challenges you had to overcome, and useful things you learned about the data in the process.
 - * If you have extra time, you might also talk about something you tried that didn't quite work out, or something that you would like to add to the report in a future update.
 - * If you tackled any of the bonus work (see last page), you may also demo that in your video. (If you don't have time to include bonus work in your 10-minute video, don't worry -- you will still get credit for that work if it is published to your Capstone_3 repository.)

- In-Class Presentations -

11. Once again, you will be delivering a short live presentation of your analysis to your classmates. This is an important “data storytelling” opportunity! To get prepared:
 - * Practice delivering your presentation in advance. Rehearse what you will share on your screen and the words that you plan to say, and make notes to help yourself stay on-track.
 - * Make sure your project files are well organized so that you can find things quickly and smoothly.
12. When it is your turn, imagine you are presenting at a regional team meeting including the director, territory managers for each state, and individual store managers.
 - * **You will have 5 minutes for your presentation.** Share your screen while presenting. Use notes as needed to help you walk through your presentation in a well-organized manner.
 - * Pick one or two of the visualizations in your report to focus on. Highlight the results that you found and why it's important to the company.
 - * Provide a brief, persuasive explanation of why this report will be valuable to the sales team to support better business decision-making.

- * (Note: For the in-class presentation, you do not need to demonstrate your underlying data model or any steps that you performed in Power Query.)

13. **When your classmates are presenting, give them your full attention.** You are encouraged to use Zoom reactions and chat during your classmates' presentations to react to things you find helpful, interesting, or insightful.

- * **It's recommended that you take notes on the projects where you have been assigned to provide peer review** (see below).

- Capstone Peer Review -

14. On presentation day, during your independent work time, you will be assigned 4 of your classmates' video projects to watch and provide a peer review via Teams using the questions provided in your Week 12 course material.

- * Peer reviews should focus on **constructive feedback**. The purpose of providing feedback is not merely to state your opinion; rather, it is to help the other person improve and refine their work.

Tips for a successful capstone project

- Make sure you implement all the requirements from page 1!
- Save all your project files to your Capstone_3 repository. Make sure you regularly commit changes using meaningful commit messages. You should have *at least one* commit every day during capstone week.
- Use good naming practices throughout your project for report pages, headings and descriptions, and text labels in visualizations. Text should be clearly readable and make sense.
- Once again it is *highly* recommended that you take some time to do a practice run or “dress rehearsal” of your presentation to check your timing, especially for your live presentation. Five minutes goes fast!
- On presentation day, have your report and any other files you plan to use open and ready to share when your turn comes up. (You can demo your Power BI report either using the desktop app or as published online, whichever you prefer.)
- As ever, make sure that the substance of your project reflects *your own independent work*. This is your opportunity to test your skills and show off what you can do!

BONUS: Capstone Project Add-ons

Once you have met the core requirements, you may extend your project with one or more of the below suggestions. (You are also welcome to explore your own ideas for how to expand the report.)

- * **OPTIONAL:** Perform an analysis that addresses other aspects of business performance. This could include analysis of top-performing stores, high- and low-selling product categories, and trends in sales of individual products.
- * **OPTIONAL:** For online shipments, EmporiUm has negotiated a deal with their shipping partner for that state. While shipping fees are not included in the Online Sales reported data, for purposes of this report they can be estimated at 2% of the sale amount. For a given month, if shipments for a state exceed \$5,000, then EmporiUm will receive half the negotiated volume discount. If shipments exceed \$10,000 in a month for that state, EmporiUm will receive the full negotiated volume discount. Create a report page that shows estimated shipping costs and calculates volume discounts by state.
- * **OPTIONAL:** Extend the customization in your report to include a custom tooltip for a visualization, a drillthrough page, custom bookmark navigation, or other interactive features.
- * **OPTIONAL:** Create a mobile version of your report that could be shared with store managers to use on a smartphone or tablet. Consider what information might be most important to highlight for mobile users and how visualizations might be best organized for readability and usability.